Pilot composition monitoring (protein and fat) for a whey stream and an opaque milk concentrate, with a comparison of measurement interfaces and first measurement concepts.

protein | fat

Over all cases (minimum per method): NIR 26 · UV-Vis 24 · MIR 23 · Fluorescence 22 · Raman 22

DECISION STATUS **Provisional feasibility assessment**

NIR leads (26/36); provisional until 'sample state constraints contact' is checked.

NIR, transmission/transflection probe · 26/36

- One Case and one reference per target. `S01 p17-18` `REF D02,D03` `R_Q1_N03`
- Clear whey → transmission; opaque concentrate → diffuse reflection; variable = both. `S01 p17-18` `REF D02,D03` `R_Q2_05`

Recommended methods

#1
26 /36
range [9–32]

NIR · transmission/transflection probe

Confidence: low | ⚠ critical question open

Q1	6 /9	Q1b	5 /9
Q2	8 /9	Q3	7 /9
Q4	2 /3		

#2
24 /36
range [9–32]

UV-Vis · transmission/transflection probe

Confidence: low · ⚠ critical question open

Q1	4 /9	Q1b	5 /9
Q2	8 /9	Q3	7 /9
Q4	0 /3		

#3
23 /36
range [8–32]

MIR · transflection probe

Confidence: low · ⚠ critical question open

Q1	4 /9	Q1b	5 /9
Q2	7 /9	Q3	7 /9
Q4	0 /3		

▼ More methods (2)

#4
22 /36
range [8–32]

Fluorescence · fluorimetric probe

Confidence: low · ⚠ critical question open

Q1	4 /9	Q1b	5 /9
Q2	6 /9	Q3	7 /9
Q4	0 /3		

#5
22 /36
range [8–32]

Raman · Raman probe

Confidence: low | ⚠ critical question open

Q1	4	/9	Q1b	5	/9
Q2	6	/9	Q3	7	/9
Q4	0	/3			

Questions for the customer

1. high `Sample · sample state constraints · contact`

   May a probe touch the product at the measuring point?

   ▼ Why

   if contact=allowed: leader ['NIR|NIR_TRANS_PROBE', 'MIR|MIR_ATR_PROBE', 'UV_VIS|UV_TRANS_PROBE']; if contact=prohibited: leader NIR

   `R_Q2_03, R_Q2_06`

2. high `Sample · sample state constraints · sampling`

   May a sample be taken out (and diluted) for the measurement, e.g. into a cuvette?

   ▼ Why

   if sampling=prohibited: leader NIR

   `R_Q2_10`

3. medium `Sample · relevant components · absorber`

   Is there a known UV/Vis absorber (chromophore) linked to the target?

   ▼ Why

   if absorber=known: leader ['NIR|NIR_TRANS_PROBE', 'UV_VIS|UV_TRANS_PROBE', 'UV_VIS|UV_ATR']

   `R_Q1_U01`

To improve accuracy, add the customer's answers to the text above and click Analyze again.

Ruled out (1)

UV_VIS · UV_LENS
This interface collects light emitted by the sample itself; the sample is not a light source. `R_Q2_02`

Decision path · factors of the leading candidate per stage, with source

Q1

MATCH **+2** Broad target-signal link; no catalog entry = UNKNOWN, never exclusion.
`S01 p4-5` `REF §3,5` `R_Q1_01`

UNKNOWN **0** Separation of the target signal from the interferences of this matrix. MATCH only where an evidence row names the page (selectivity_source); no spectra in the materials = UNKNOWN, never a MISMATCH and never 'no interferences'. `S01 p5,10-11` `R_Q1_04`

UNKNOWN **0** Generic row: a known matrix background must not cover the target signal. Unknown overlap = UNKNOWN. Replaced by the specific rows R_Q1_R02 (Raman fluorescence) and R_Q1_M03 (MIR water). Soft only; a hard exclusion needs a proven overlap (R_Q1_02). `S01 p5,11` `REF §5` `R_Q1_05`

UNKNOWN **0** Invalid old calibration does not exclude the physical method; a new calibration plan is mandatory. `REF D01 risks, §5` `S01 p10` `R_Q1_06`

UNKNOWN **0** Independent validation material must be established, not assumed.
`REF D01/D02 experiment` `S01 p5,10` `R_Q1_07`

UNKNOWN **0** Particle, density, temperature, texture variation must be in the matrix calibration.
`S01 p10,15,17` `REF D01,D02` `R_Q1_N02`

MATCH **+2** One Case and one reference per target. `S01 p17-18` `REF D02,D03` `R_Q1_N03`

Q1b

UNKNOWN **0** LOD/LOQ only from a scoped CapabilityProfile; the materials give none.
`spec 04 §1` `REF §3,5` `R_Q1B_02`

UNKNOWN **0** Achieved error vs allowed error needs a comparable metric; none in the materials.
`REF D01/D02 experiment` `spec 04` `R_Q1B_04`

MATCH **+1** A laboratory is not a known reference uncertainty. `REF D01,D02` `H01` `R_Q1B_05`

UNKNOWN **0** End-to-end latency needs a profile; spectral exposure alone is not the latency.
`spec 04 §1` `R_Q1B_06`

UNKNOWN **0** Both range ends must be inside the calibration/detector scope.
`REF D01,D02` `S01 p16,18` `R_Q1B_08`

UNKNOWN **0** Qualitative sensitivity rating: context only, never a score. `XLSX sensitivity + notes` `R_Q1B_10`

Q2

MATCH **+1** Documented Method-interface edge (S01 Table 1). `XLSX` `S01 p4-6` `R_Q2_01`

MATCH **+1** A form missing from the list is UNKNOWN; MISMATCH only for a constructive conflict (solid vs liquid-only cell). `XLSX` `CL10` R_Q2_02

UNKNOWN **0** Excludes only this pair; other interfaces stay. Taking a sample out is R_Q2_10 (one cause, one rule). `S01 p5,10` R_Q2_03

MATCH **+2** Turbidity without a spectrum is a risk, not a global exclusion. Check at maximum turbidity. `S01 p16,18` R_Q2_04

MATCH **+2** Clear whey → transmission; opaque concentrate → diffuse reflection; variable = both. `S01 p17-18` `REF D02,D03` R_Q2_05

UNKNOWN **0** A surface or local signal is not the bulk composition of a heterogeneous stream without a sampling plan. `S01 p8,13-14,17-18` `REF D08` R_Q2_07

UNKNOWN **0** Particle/bubble/surface variation must be covered by a sampling or averaging plan. Our recommendation to average is not a plan (no MATCH from it). `REF D01,D02` `S01 p15,17` R_Q2_08

Q3

UNKNOWN **0** No numeric limit in the materials. Hot extrusion does not exclude MIR. `S02 p14` `REF D07` R_Q3C_01

UNKNOWN **0** Fouling + contact interface needs a feasible cleaning/maintenance plan; a promise is not a match. `REF D02,D07` R_Q3C_04

MATCH **+2** Preference is soft; a lab interface stays valid for the feasibility experiment. `REF §3` R_Q3C_05

UNKNOWN **0** Do not invent certification. `design comparator` R_Q3C_06

UNKNOWN **0** Long silica cables are documented; no universal allowed length. `S02 p6-10,18` `S01 p15` R_Q3C_07

MATCH **+1** Generic family is a soft concept check; a real cable with proven mismatch is R_Q3P_03. `S02 p6` `REF §5` R_Q3C_08

Risks and assumptions

high The critical open point is whether a probe may touch the product and whether samples may be taken out. All answers change the leading method, so the NIR lead is conditional.

high All top candidates have low confidence, provisional status and wide score intervals. NIR has only weak evidence of closeness (evidence score 2) and UV-Vis and MIR have none, so no method has been shown in an AFS application for this case. The scores are not probabilities, and NIR's lead over UV-Vis and MIR (26 vs 24 vs 23) is small.

high The case turbidity feature says 'clear', which fits only the whey. The concentrate is opaque. One probe geometry may not serve both streams, so plan separate interfaces and calibrations.

medium Concentration ranges, temperature, fouling, pressure, instrument distance and hazardous-area status are all unknown. Fat and protein levels in dilute whey may be low enough for water bands to hide the signal in NIR and MIR.

medium UV-Vis rests on an absorber linked to protein or fat that is not established, and it suffers from scattering in the opaque concentrate. It is a baseline for comparison, not a favourite.

medium MIR depends on a short, costly fiber link whose maximum length is not given in the AFS materials. If the analyzer must be far from the line, MIR may be impractical. Placing the analyzer at the line and sending the result digitally would keep MIR open.

medium The laboratory protein and fat results are the only reference, and their uncertainty and independence are unknown. This limits what the calibration can show.

medium Fat and protein films can foul probe windows in dairy lines, and cleaning cycles and temperature are unknown. This could hit the transflection gap or window and may require cleaning or a bypass.

low Raman and fluorescence score 22/36 and are only just below MIR (23). They are not in the top three and are not in the experiment. If the contact question or the first results go badly, they would need a second look.

First feasibility experiment

Test on static samples whether NIR, UV-Vis and MIR fiber probes can measure protein and fat in both the clear whey and the opaque milk concentrate, compared with the laboratory reference.

Samples

About 40-60 samples per stream (whey and concentrate), split into at least 2 calibration batches and 2 independent validation batches. Final numbers to be agreed with the customer. — Use real production samples plus designed samples. Spike or dilute to vary protein and fat independently, so the two do not correlate with each other or with total solids and water. Cover the expected concentration range, temperature range and turbidity from clear whey to the opaque concentrate. Include several production batches or recipes.

Reference method

The customer's laboratory protein and fat methods, with duplicate reference measurements per sample to estimate reference repeatability.

Methods compared

NIR, UV-Vis, MIR

Interferents

fat vs protein correlation, total solids / water content, lactose and other dissolved components, temperature, turbidity and particle scattering, fouling films on the probe window

Duration

About 4-6 weeks for the laboratory study including sample collection and reference analysis. An in-line pilot campaign would follow separately and take longer.

1. 1. Clarify the target and the measurement point: contact allowed, sampling allowed, one or two streams, expected ranges, accuracy need, distance to the analyzer.
2. 2. Collect representative whey and concentrate samples from several batches and send aliquots to the laboratory for the reference values.
3. 3. Prepare spiked and diluted series so protein and fat vary independently. Record temperature for every sample.
4. 4. Measure every sample with the NIR (transflection/transmission, diffuse reflection for the concentrate), UV-Vis and MIR fiber probes under the same conditions.
5. 5. Build a calibration for each method and stream on the calibration batches. Keep single-method baselines.
6. 6. Validate on independent batches that played no part in the calibration, split by batch and not by neighbouring spectra of one sample.
7. 7. Compare methods with the agreed criterion and decide whether to go on to an in-line pilot campaign. Check probe fouling in a short flow-through or recirculation trial.

Success criterion: On independent validation batches, for each target and stream, the prediction error (RMSEP) is within a limit agreed with the customer beforehand, and the bias against the laboratory reference is not significantly different from zero. The limits are agreed once the reference repeatability is known.

Parameters and attributes · 38 Facts

Facts

Value · unit · status. Hover a row for the customer's words. Edit → Recalculate.

Fact	Value	Unit	Status
MeasurementTask · measurement goal	quantify		inferred
MeasurementTask · target	protein, fat		reported „…“
MeasurementTask · target definition	defined		reported „…“
MeasurementTask · spatial scope	bulk		inferred
MeasurementTask · sampling plan	—		unknown
MeasurementTask · reference method	available		reported „…“
MeasurementTask · performance requirements	—		unknown

Fact	Value	Unit	Status
MeasurementTask · time requirements	—		unknown
MeasurementTask · target range	—		unknown
MeasurementTask · business goal	pilot		reported „…“
Sample · matrix	dairy		reported „…“
Sample · phase	liquid		inferred
Sample · physical form	solution		inferred
Sample · optical properties · opacity	clear		reported „…“
Sample · optical properties · fluorescence	—		unknown
Sample · heterogeneity	—		unknown
Sample · variability · formulation	—		unknown
Sample · variability · physical	—		unknown
Sample · sample state constraints · contact	—		unknown
Sample · sample state constraints · sampling	—		unknown
Sample · relevant components · fluorophore	—		unknown
Sample · relevant components · absorber	—		unknown
ProcessContext · deployment preference	—		unknown
ProcessContext · temperature range	—		unknown
ProcessContext · temperature c	—	degC	unknown
ProcessContext · operation constraints · fouling	—		unknown
ProcessContext · operation constraints · hazardous	—		unknown

Fact	Value	Unit	Status
ProcessContext · instrument distance	—		unknown
ProcessContext · assets · fiber	—		unknown
ProcessContext · assets · fiber profile	—		unknown
ProcessContext · assets · must reuse	—		unknown
ProcessContext · assets · calibration scope	—		unknown
MeasurementTask · reference method · independent	—		unknown
ProcessContext · assets · method	—		unknown
CustomerRequest · justification	—		unknown
CustomerRequest · proxy allowed	—		unknown
CustomerRequest · asks	assessment		reported „…“
CustomerRequest · combination to keep calibration	—		unknown